Eva Janoušková

Health system modeller with 6 years of experience bridging mathematical modelling, health economics, and data science to drive evidence-based policy. Passionate about real-world impact and equitable public health, recognising patient, system, and resource barriers.

✉ *janouskova.eva.012@gmail.com*

LinkedIn | *Google Scholar* | *GitHub*
ResearchGate | *ORCID* | *ResearcherID*

Work Experience

08/2022 – 10/2025 **Research Fellow in Health System Modelling**, *University College London, Institute for Global Health*, London, United Kingdom.
Developed and integrated malnutrition and medical equipment modules within simulation individual-based model (IBM) of Malawian health system (TLO model). Led the end-to-end research lifecycle for the acute malnutrition – including design, Python-based implementation, parametrisation, calibration, HPC execution, cost-effectiveness policy evaluation, uncertainty quantification, sensitivity analysis, and interpretation. Finalised the contraception module and assessed the impacts and costs of modern contraceptive uptake campaigns. Collaborated on a range of multidisciplinary studies modelling disease dynamics (evaluating budget allocation, investment strategies, and intervention cost-effectiveness). Contributed to grant applications and provided rigorous peer code and research reviews. Managed stakeholder relationships with the Malawian Ministry of Health (MoH) to inform the model and translate findings into evidence-based policy recommendations. Engaged in postgraduate research supervision and teaching (UCL, KUHeS).

12/2019 – 06/2022 **Research Fellow in Mathematical Modelling**, *University of Surrey, School of Veterinary Medicine*, Guildford, United Kingdom.
Developed a stochastic geospatial network model in R to simulate zoonotic rabies spillover in Brazil: cleaned, validated, and integrated ecological niche datasets, mapped spatial risk, evaluated vaccination strategies, and visualised outcomes for the Brazilian MoH. Adapted an open-source individual-environment simulation model of schistosomiasis transmission in Julia to evaluate WASH interventions. Calibrated both models using Approximate Bayesian Computation (ABC) via custom R scripts and Julia packages, run simulations on HPC clusters with Docker. Published a guide outlining future research directions for sustainable disease control programmes. Contributed to a co-infection data analysis in the Philippines. Collaborated across international research teams, mentored junior researchers, contributed statistical input and peer reviews for grant proposals, and delivered postgraduate teaching and PhD project supervision.

09/2010 – 05/2019 **External Lecturer**, *Masaryk University, Studies Section, Centre for Students with Special Needs*, Brno, Czech Republic.
Adapted and delivered Bachelor/Master lectures and seminars across multiple faculties. Subjects: Theoretical with applications (differential and integral calculus, combinatorics, advanced probability, graph theory), Practical with demonstrations in R, STATISTICA, and SAS (probability, statistical modelling).

Education

2019 **Doctoral degree**, *Masaryk University, Faculty of Science*, Brno, Czech Republic.
Field "Applied Mathematics", specialisation "**Probability, Statistics and Mathematical Modelling**", thesis "Mathematical modeling / Impacts of trade-offs on dynamics of sterilizing infections".
Developed population-level models to investigate how host life-history trade-offs impact the dynamics of sterilising infections, utilising MATLAB for model analyses and visualisation. Investigated the potential of pathogens as biocontrol agents and the evolution of pathogen sterilising strategies. Teaching: Delivered seminars for BSc students at Faculty of Informatics (Linear Models, Differential and Integral Calculus, and Statistics I); reviewed and marked BSc theses at the Faculty of Science.

2012 **Master's degree**, *Masaryk University, Faculty of Science*, Brno, Czech Republic.
Field "Applied Mathematics", specialisation "**Statistics and Data Analysis**", thesis "Singular value decomposition and its usage". (MATLAB-based implementation.)

2009 **Bachelor's degree**, *Masaryk University, Faculty of Science*, Brno, Czech Republic.
Field "Applied Mathematics", specialisation "**Statistics and Data Analysis**", thesis "Combinatorics".

Additional Experience & Leadership

Peer review	PLOS ONE & PLOS Neglected Tropical Diseases
Co-supervision	PhD rotation project (2021), student is currently pursuing a career in mathematical modelling
Supervision	MSc dissertation (2024/2025), received second best mark in the cohort
Volunteering	Math Tutor at Action Tutoring, a national education charity supporting pupils from disadvantaged backgrounds

Skills

Modelling	IBMs & Compartmental models, Stochastic & Deterministic simulation, Markov, State-transition, & Time-to-event modelling, Network modelling, Regression
Economics	Cost-Effectiveness & Utility Analysis (CEA/CUA), Allocative efficiency & budget allocation, DALYs & QALYs, ICER threshold analysis, CE frontier, INHB & INMB metrics, Discounting (costs & health outcomes)
Data Science	Bayesian inference (ABC), Parametrisation & Calibration, Chained multi-stage simulation pipelines, Uncertainty quantification (stochasticity, probabilistic and multi-way sensitivity analysis), Geospatial risk mapping, Dimensionality reduction (PCA, SVD)
Data Analysis	SQL, API data extraction, Data wrangling (Pandas, NumPy, Tidyverse), Exploratory Data Analysis (EDA), Missing data imputation, ETL (Extract, Transform, Load) pipelines
Languages	Python, Julia, R, MATLAB, NetLogo
DevOps	Version control & Code review (Git/GitHub), Unit testing (Pytest), Copilot, Containerisation & Environments (Docker, Conda/Venv), High-performance computing (HPC: SGE, HTCondor)
Reporting	LaTeX, ConTeXt, Word, PowerPoint, Interactive Excel dashboards, Visualisation (Matplotlib, Seaborn, ggplot2), STATISTICA

Courses & Professional Development

06/2025	Introduction to qualitative analysis: Interviewing; UCL (UK)
04/2025	Supervising Masters Projects and Dissertations; UCL (UK)
02/2025	Resilient Leadership in Action; King's College London (UK)
06/2024	Applying for an Early Career Research Fellowship; UCL (UK)
03/2020	Advanced Bayesian Disease Mapping course; Medical University of South Carolina (USA)
07/2014	6th Summer Institute in Statistics and Modeling in Infectious Diseases; University of Washington (USA) / 3 modules: Mathematical Models of Infectious Disease; Infectious Diseases Immunology and Within-Host Models; Stochastic Epidemic Models with Inference. **Awarded full scholarship.**
09/2013	9th Summer School of Mathematical Biology, Stochastic Modelling in Epidemiology (CZ)

Publications

submitted Bhatia S., Y. M. S. Althobaity, L. A. C. Chapman, J. Chemutai, C. D. S. Estadilla, L. Gansberger, **E. Janoušková**, P. N. Karega, T. Leng, R. McCabe, R. E. Murray-Watson, T. M. Naidoo, M. Pritchard, B. She, T. B. Hallett, A. Petherick, P. Nouvellet, A. Cori, and A. Mavrodaris. **Better data for better public health decision-making: quantifying the intended and unintended impacts of public service modifications**.

2026 Perinpakumar A., B. She, T. Mangal, S. Mohan, M. Chalkley, T. Colbourn, J. H. Collins, M. M. Graham, **E. Janoušková**, D. Nkhoma, P. D. Twea, A. N. Phillips, P. Revill, A. U. Tamuri, J. Mfutso-Bengo, T. B. Hallett, and M. Molaro. **An estimation of the health-cost of unfilled medical positions in Malawi: A Thanzi La Onse Mathematical Modelling study**. *medRxiv*. https://doi.org/10.64898/2026.05.25.26353761.

2026 Mangal T. D., T. Colbourn, A. Philips, J. Mfutso-Bengo, P. Mphamba, S. Mohan, R. Murray-Watson, D. Nkhoma, **E. Janoušková**, B. She, P. Revill, and T. Hallett. **Linking transmission dynamics and economic evaluation to assess schistosomiasis programme strategies across districts in Malawi**. *medRxiv (submitted: PLOS NTDs)*. https://doi.org/10.64898/2026.05.06.26352519.

2026 Mohan S., N. Chagoma, S. Walker, C. A. Arega, M. Chalkley, J. Collins, E. Connolly, T. Colbourn, **E. Janoušková**, T. D. Mangal, G. Manthalu, J. Mfutso-Bengo, M. Molaro, D. Nkhoma, A. Phillips, L. Sharma, B. She, W. Tafesse, P. D. Twea, P. Revill, and T. B. Hallett. **Estimating system-wide healthcare costs using a health system model: Application to the *Thanzi La Onse* model of Malawi**. *Appl Health Econ Health Policy*. https://doi.org/10.1007/s40258-026-01030-w.

2026 **Janoušková E.**, I. Li Lin, E. Mnjowe, W. Mulwafu, E. Connolly, S. Mohan, D. Nkhoma, A. Seal, J. Mfutso-Bengo, M. Chalkley, J. H. Collins, T. D. Mangal, P. N. Mphamba, R. E. Murray-Waston, J. Phuka, A. U. Tamuri, A. N. Phillips, P. Revill, T. B. Hallett, and T. Colbourn. **Cost-effectiveness of addressing constraints in childhood acute malnutrition management in Malawi using the *Thanzi La Onse* health system simulation framework**. *medRxiv (submitted: PLOS Med.)*. https://doi.org/10.64898/2026.03.05.26347696.

2025 Murray-Watson R. E., M. Molaro, R. J. Murray-Watson, S. Mohan, B. She, T. Mangal, J. H. Collins, S. Bhatia, **E. Janoušková**, and T. B. Hallett. **The impact of precipitation on ANC service utilisation and healthcare access in Malawi**. *Sci. Rep.*, 15(1):37896. ISSN 2045-2322. https://doi.org/10.1038/s41598-025-21645-8.

2025 Murray-Watson R. E., M. Molaro, T. D. Mangal, S. Mohan, B. She, S. Bhatia, J. H. Collins, **E. Janoušková**, T. Colbourn, A. N. Phillips, and T. B. Hallett. **Modelling the long-term demographic and epidemiological trends in Malawi**. *medRxiv*. https://doi.org/10.1101/2025.09.25.25336670.

2025 She B., S. Mohan, R. E. Murray-Waston, S. Bhatia, M. Chalkley, T. Colbourn, J. H. Collins, E. Connolly, **E. Janoušková**, D. Nkhoma, P. Revill, A. U. Tamuri, P. D. Twea, T. D. Mangal, J. Mfutso-Bengo, T. B. Hallett, and M. Molaro. **Health impacts of expanding different health workforce cadres under a limited budget in Malawi**. *medRxiv*. https://doi.org/10.1101/2025.09.13.25335695.

2025 Collins J. H., W. Tafesse, P. Chitsulo, T. B. Hallett, **E. Janoušková**, J. Mfutso-Bengo, E. Mnjowe, S. Mohan, W. Mulwafu, D. Nkhoma, B. She, M. Suarez, and T. Colbourn. **Healthcare service user-reported quality of care in Malawi: a national multi-facility cross-sectional study**. *medRxiv*. https://doi.org/10.1101/2025.09.10.25335306.

2025 Molaro M., P. Revill, M. Chalkley, S. Mohan, T. D. Mangal, T. Colbourn, J. H. Collins, M. M. Graham, W. Graham, **E. Janoušková**, G. Manthalu, E. Mnjowe, W. Mulwafu, R. E. Murray-Watson, P. D. Twea, A. N. Phillips, B. She, A. U. Tamuri, D. Nkhoma, J. Mfutso-Bengo, and T. B. Hallett. **The potential impact of declining development assistance for health on population health in Malawi: A modelling study**. *PLOS Med.*, 22(8):e1004488. ISSN 1549-1676. https://doi.org/10.1371/journal.pmed.1004488.

2025 Li Lin I., E. D. McCollum, E. Buckley, J. H. Collins, M. M. Graham, **E. Janoušková**, C. King, N. Lufesi, T. D. Mangal, J. Mfutso-Bengo, E. Mnjowe, S. Mohan, M. Molaro, D. Nkhoma, H. Nsona, A. Rothkopf, B. She, L. Smith, A. U. Tamuri, P. Revill, V. Cambiano, A. N. Phillips, T. B. Hallett, and T. Colbourn. **The impact and cost-effectiveness of pulse oximetry and oxygen on acute lower respiratory infection outcomes in children in Malawi: a modelling study**. *The Lancet Glob. Health*, 13(8): e1466–e1475. ISSN 2214109X. https://doi.org/10.1016/S2214-109X(25)00202-5.

2025 Mangal T. D., M. Molaro, D. Nkhoma, T. Colbourn, J. H. Collins, **E. Janoušková**, M. M. Graham, I. Li Lin, E. Mnjowe, T. E. Mwenyenkulu, S. Mohan, B. She, A. U. Tamuri, P. D. Twea, P. Winskill, A. Phillips, J. Mfutso-Bengo, and T. B. Hallett. **Modelling health outcomes of a decade of HIV, malaria and tuberculosis initiatives, Malawi**. *Bull. World Health Organ.*, 103(5):304–315. ISSN 0042-9686. https://doi.org/10.2471/BLT.24.292439.

2025 Mangal T. D., S. Mohan, M. Molaro, J. Collins, T. Colbourn, **E. Janoušková**, R. Murray-Watson, D. Nkhoma, A. Phillips, B. She, P. D. Twea, S. Walker, P. Revill, and T. B. Hallett. **System-wide investments enhance HIV, TB and malaria control in Malawi and deliver greater health impact**. *medRxiv*. https://doi.org/10.1101/2025.04.29.25326667.

2025 Collins J. H., H. Allott, W. Ng'ambi, I. Li Lin, M. Giordano, M. M. Graham, **E. Janoušková**, F. Kachale, K. Kawaza, T. D. Mangal, J. Mfutso-Bengo, E. Mnjowe, S. Mohan, M. Molaro, D. Nkhoma, P. Revill, A. Rodger, B. She, A. U. Tamuri, C. J. Tann, P. D. Twea, V. Cambiano, T. B. Hallett, A. N. Phillips, and T. Colbourn. **An individual-based modelling study estimating the impact of maternity service delivery on health in Malawi**. *Nat. Commun.*, 16(1):3925. ISSN 2041-1723. https://doi.org/10.1038/s41467-025-59060-2.

2025 Hallett T. B., T. D. Mangal, A. U. Tamuri, N. Arinaminpathy, V. Cambiano, M. Chalkley, J. H. Collins, J. Cooper, M. S. Gillman, M. Giordano, M. M. Graham, W. Graham, I. Hawryluk, **E. Janoušková**, B. L. Jewell, I. Li Lin, R. Manning Smith, G. Manthalu, E. Mnjowe, S. Mohan, M. Molaro, W. Ng'ambi, D. Nkhoma, S. Piatek, P. Revill, A. Rodger, D. Salmanidou, B. She, M. Smit, P. D. Twea, T. Colbourn, J. Mfutso-Bengo, and A. N. Phillips. **Estimates of resource use in the public-sector health-care system and the effect of strengthening health-care services in Malawi during 2015–19: a modelling study (Thanzi La Onse)**. *The Lancet Glob. Health*, 13(1):e28–e37. ISSN 2214-109X. https://doi.org/10.1016/S2214-109X(24)00413-3.

2024 Mangal T. D., S. Mohan, T. Colbourn, J. H. Collins, M. M. Graham, A. Jahn, **E. Janoušková**, I. Li Lin, R. M. Smith, E. Mnjowe, M. Molaro, T. E. Mwenyenkulu, D. Nkhoma, B. She, A. U. Tamuri, P. Revill, A. N. Phillips, J. M. Mfutso-Bengo, and T. B. Hallett. **Assessing the effect of health system resources on HIV and tuberculosis programmes in Malawi: a modelling study**. *The Lancet Glob. Health*. https://doi.org/10.1016/S2214-109X(24)00259-6.

2024 Molaro M., S. Mohan, B. She, M. Chalkley, T. Colbourn, J. H. Collins, E. Connolly, M. M. Graham, **E. Janoušková**, I. Li Lin, G. Manthalu, E. Mnjowe, D. Nkhoma, P. D. Twea, A. N. Phillips, P. Revill, A. U. Tamuri, J. Mfutso-Bengo, T. D. Mangal, and T. B. Hallett. **A new approach to Health Benefits Package design: an application of the Thanzi La Onse model in Malawi**. *PLOS Comput. Biol.*, 20(9). ISSN 1553-7358. https://doi.org/10.1371/journal.pcbi.1012462.

2024 Mohan S., T. D. Mangal, T. Colbourn, M. Chalkley, Ch. Chimwaza, J. H. Collins, M. M. Graham, **E. Janoušková**, B. Jewell, G. Kadewere, I. L. Lin, G. Manthalu, J. Mfutso-Bengo, E. Mnjowe, M. Molaro, D. Nkhoma, P. Revill, B. She, R. Manning Smith, W. Tafesse, A. U. Tamuri, P. Twea, A. N. Phillips, and T. B. Hallett. **Factors associated with medical consumable availability in level 1 facilities in Malawi: a secondary analysis of a facility census**. *Lancet Glob. Health*. https://doi.org/10.1016/S2214-109X(24)00095-0.

2024 She B., T. D. Mangal, A. Y. Adjabeng, T. Colbourn, J. H. Collins, **E. Janoušková**, I. L. Lin, E. Mnjowe, S. Mohan, M. Molaro, A. N. Phillips, P. Revill, R. Manning Smith, P. D. Twea, D. Nkhoma, G. Manthalu, and T. B. Hallett. **The changes in health service utilisation in Malawi during the COVID-19 pandemic**. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0290823.

2023 Colbourn T., **E. Janoušková**, I. Li Lin, J. Collins, E. Connolly, M. Graham, B. Jewel, F. Kachale, T. Mangal, G. Manthalu, J. Mfutso-Bengo, E. Mnjowe, S. Mohan, M. Molaro, W. Ng'ambi, D. Nkhoma, P. Revill, B. She, R. Manning Smith, P. Twea, A. Tamuri, A. Phillips, and T. B. Hallett. **Modeling contraception and pregnancy in Malawi: a *Thanzi La Onse* mathematical modeling study**. *Stud. Fam. Plan.* https://doi.org/10.1111/sifp.12255.

2023 Hallett T., A. Phillips, A. Tamuri, T. Colbourn, J. Collins, I. Hawryluk, B. Jewell, I. L. Lin, T. Mangal, R. Manning-Smith, S. Mohan, W. Ng'ambi, B. She, M. Gillman, M. Giordano, M. Graham, E. Mnjowe, S. Piatek, J. Cooper, D. Salmanidou, W. Graham, M. Molaro, and **E. Janoušková**. **The Thanzi La Onse Model**. *zenodo*. https://doi.org/10.5281/zenodo.10144015.

2023 **Janoušková E.**, J. Rokhsar, M. Jara, M. Entezami, D. L. Horton, R. A. Dias, G. Machado, and J. M. Prada. **Quantifying spillover risk with an integrated bat-rabies dynamic modeling framework**. *Transbound. Emerg. Dis.* https://doi.org/10.1155/2023/2611577.

2022 **Janoušková E.**, J. Clark, O. Kajero, S. Alonso, P. H. L. Lamberton, M. Betson, and J. M. Prada. **Public health policy pillars for the sustainable elimination of zoonotic schistosomiasis**. *Front. Trop. Dis.* https://doi.org/10.3389/fitd.2022.826501.

2022 Kajero O., **E. Janoušková**, E. A. Bakare, V. Belizario, B. Divina, A. J. Alonte, S. M. Manalo, V. G. Paller, M. Betson, and J. M. Prada. **Co-infection of intestinal helminths in humans and animals in the Philippines**. *Trans. R. Soc. Trop. Med. Hyg.*, page trac002. ISSN 0035-9203. https://doi.org/10.1093/trstmh/trac002.

2020 **Janoušková E.** and L. Berec. **Fecundity-longevity trade-off, vertical transmission and evolution of virulence in sterilizing pathogens**. *Am. Nat.*, 195:95–106. https://doi.org/10.1086/706182.

2018 **Janoušková E.** and L. Berec. **Host-pathogen dynamics under sterilizing pathogens and fecundity-longevity trade-off in hosts**. *J. Theor. Biol.*, 450:76–85. https://doi.org/10.1016/j.jtbi.2018.04.017.

2017 Berec L., **E. Janoušková**, and M. Theuer. **Sexually transmitted infections and mate-finding Allee effects**. *Theor. Popul. Biol.*, 114:59–69. https://doi.org/10.1016/j.tpb.2016.12.004.

London, UK, 4th September 2026